

Selected Work

- Designed and engineered the **first remote MCP server infrastructure** in Amazon Devices & Services Finance, onboarding 18+ tools (called 5,500+ times annually) that read sensitive financial data through **authenticated, permissioned access**. [AWS blog under review]
- Engineering Lead for UMass Amherst’s Alexa Prize team - Marunabot v2 (\$250k awarded, top 10 of 100+ teams, #1 in semifinals). Built the dialog manager on Amazon’s CoBot framework, orchestrating **2 intent classifiers** and ranking **21 parallel LLM response generators**. Designed **auto-scaling LLM serving on AWS hitting 7 TPS** despite sequential per-generator latency. Amazon shipped the bot to all US Alexa devices via ”Alexa, let’s chat” - 66k+ conversations in 3 months. NSF (\$570k grant) + Amazon backed. [Paper on Amazon Science].
- Started Poll Plus, open source polling app for Rocket.Chat. Architected **Live Polls with persistent storage for scheduled deployment**, and built an **open-source word cloud API to meet Credit Suisse’s air-gapped constraints**. Credit Suisse and Google funded; piloted to 10k employees across Asia. First app packaged in Rocket.Chat’s official docs (12M user base). [Rocket.Chat Engineering Blog] [GitHub Repository].
- Built and open-sourced a **YOLOv3-based** license plate recognition framework for Indian vehicles, trained on 2,500+ manually collected images. #1 globally for the task on GitHub by popularity, top 0.7% of 1,000+ LPR repos. [GitHub Repository]

Experience

SDE I → SDE II, Applied AI

Amazon

June 2024 - Present | Sunnyvale, CA

- Built the core harness for Ask Rino, agentic AI companion for Amazon Devices & Services Finance, including **LLM-routing, websocket streaming** to the frontend, **RAG over finance documents with citations**, internal search over Amazon resources and **pre-signed URL generation** for secure file uploads on AWS Bedrock AgentCore.
- Extended the MCP protocol to a **third transport** that used **FastMCP as proxy** and **browser native messaging** to communicate with local MCP servers through the Ask Rino browser extension (Claude Cowork equivalent).
- Developed strategies for **short-term and long-term memory extraction and consolidation** from session messages using Agent-Core Memory. Strategies spanned across user preferences, session summaries, episodic and custom lessons retrieved based on **relevance and top-k ranking with temporal ordering**.
- Impact: 400+ Amazon finance managers use Ask Rino to analyze and create artifacts in their browser and on local files. 41,000+ conversations annually, VP-level Kingpin goal cited by Amazon CFO Brian Olsavsky.

Open Source Engineer

Google Summer of Code

Oct 2020 – August 2022 | GitHub

- Selected to Google Summer of Code out of 50,000+ applicants (sub-2.5% acceptance rate) as a Contributor (2021) and a Mentor (2022) for Rocket.Chat.
- Contributor: Rocket.Chat **core engine** (fixes across 3 releases), Opensource Contribution Leaderboard (**new APIs, dockerization**), RC4Community (code + collab platform, founder, full-stack), Poll Plus (founder, full-stack).
- Mentor: EmbeddedChat (embed Rocket.Chat anywhere, top 10 most popular), GitHub App (GitHub app for Rocket.Chat).

Honors and Activities

- Self-filed and approved **EB-1A** (Einstein green card) in Software Engineering and Applied AI (2025), no lawyer took the case.
- Community Contributor of the Month, Rocket.Chat (Apr 2022), for contributing over 500,000 lines of code to Rocket.Chat’s open-source repositories - highest that year.
- Best Innovation Award, for paper on fast hand-pose recognition using 3D CNNs at ICICC 2020 (803 submissions, 29 countries).

Education

University of Massachusetts Amherst

MS Computer Science | May 2024 | GPA: 3.87/4.00

Research: Alexa Prize - CIIR (US #2 in Information Retrieval), CarbonCast - LASS (US #3 in HPC), OpenReview at IESL.

Coursework: Distributed Operating Systems, Information Retrieval, Advanced NLP

Indian Institute of Information Technology (IIIT) Pune

BS Computer Science | June 2021 | GPA: 9.33/10.00

(Department Rank 1, Institute Merit Award)

Coursework: Data Structures and Algorithms, Cloud Computing, Machine Learning, OOP